

NETWORK INFRASTRUCTURE PRACTICAL MANUAL

Module: Network Infrastructure

Programme: Information Technology / Networking Support

Practical Assessments: 9

Total Marks: 100

Lab Size: 10 Student Workstations

1. MODULE OVERVIEW

This practical module equips students with the knowledge and skills required to design, install, configure, manage, test, and troubleshoot network infrastructure in a modern enterprise environment.

The practical exercises progress from basic cabling and hardware installation to Cisco router and switch configuration, VLAN implementation, network services, and troubleshooting.

2. AVAILABLE EQUIPMENT

Structured Cabling Equipment

Item	Quantity
Crimping Tools	4
Punchdown Tools	4
Patch Panels	4
Network Patch Boxes	13
0.5m Fly Leads	13
1m Fly Leads	13
Network Cable (500m)	4 Rolls

Cisco Equipment

Item	Quantity
Cisco Console Cables	4
Cisco 1G41 Routers	2
Cisco ISR 4331 Routers	2
Cisco Catalyst 3560 Switches	2

Network Infrastructure Equipment

Item	Quantity
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24-Port Switches	4
24-Port Patch Panels	4
4U Switch Cabinets	4

Computer Equipment

Item	Quantity
Lenovo Desktop PCs	10
LG Screens	10
Keyboard s Mouse Sets	10

Printing Equipment

Item	Quantity
HP LaserJet 600 M602 Printer	1
HP Color LaserJet 5500DN Duplex Printer	1
HP Colour LaserJet Professional CP5225N Printer	1

Toolkits

Item	Quantity
SPROTEK 58-Piece Computer Technician Toolkit	2
SPROTEK 10-Piece Network Installation Toolkit	2

Classroom Equipment

Item	Quantity
Projector	1
Projector Mounting	1
Whiteboard	1

3. LEARNING OUTCOMES

Upon successful completion of this module, students will be able to:

1. Safely use networking tools and equipment.
2. Construct Ethernet cables using industry standards.
3. Install and terminate patch boxes and patch panels.
4. Build structured cabling systems.
5. Install and manage switch cabinets.
6. Access and configure Cisco IOS devices.

7. Configure Cisco switches and routers.
 8. Implement VLANs and network segmentation.
 9. Configure shared network resources.
 10. Troubleshoot enterprise network faults.
 11. Document network installations professionally.
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PRACTICAL 1: NETWORK TOOLS AND SAFETY

Objective

Identify networking tools and apply safe working procedures.

Activities

1. Identify networking tools.

Punchdown tool

Crimping tool

Toolkit

Fly leads

Cable tester

2. Explain the use of each tool.

Punchdown tool- This tool will push conductors into IDC slots making sure they not damaged.

Crimping tool- This tool ensures secure RJ 45 connector attachment.

Toolkit- It provides safe handling equipment like screwdrivers for maintenance and installations.

Fly leads- It is cables that provide flexible connections between patch panel and switch.

Cable tester- This tool checks for open circuits and mis wiring.

3. Demonstrate safe handling.

Follow health and safety rules. For example, don't leave wires exposed and make sure all cables are secured.

Make sure the power is switched off before you touch any active network devices.

Always put on safety glasses when you cut or crimp cables and make sure that you maintain a clean workstation to prevent any accidents from occurring.

4. Complete a tool inventory sheet.

Tool	What it does	How to use it safely
Crimping Tool	Puts the plug RJ 45 onto a network cable.	Keep fingers away from the jaws, cut cables neatly and wear safety glasses.
Punchdown Tool	Pushes wires into patch panels or wall boxes.	Press firmly but carefully don't let it slip, hold your wrist properly
Cable Tester	Checks if the cable is wired correctly.	Only test cables that are not live and handle gently.
Patch Panel	A board where many cables are connected.	Mount it properly, label each port and keep cables tidy.
Patch Box	The wall socket where you plug in a cable.	Wire it using the T568B order, cover it properly and label it.
Fly Leads	Small cables to connect devices.	Don't bend them too much and check plugs before use.
Network Cable	The main cable used for building networks.	Store neatly and cut with the right tool.
Cisco Console Cable	Connects a computer to a Cisco device for setup.	Don't bend too hard and unplug carefully.
Toolkit	A set of screwdrivers, pliers and other tools.	Keep tools in order and use the right tool for the job

5. Apply health and safety procedures.

Do not use any material that is faulty or damaged and always check if there are no dangers like loose cables that can harm you. Make sure that you prevent any danger by wearing personal protective equipment.

Deliverable

Completed Tool Identification Worksheet.

Assessment Rubric (10 Marks)

Criteria	Marks
Tool identification	4
Understanding tool functions	2
Safe tool usage	2
Worksheet completion	2
Total	10
